



DECK# Design Notes

Design Decision Log

Version: 1

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DC-01 — System name: DECK

Decision: the system is named DECK#. **Alternative considered:** Deckpack. **Rationale:** DECK# (from Greek: language, vocabulary) reflects the nature of the system — a shared vocabulary for communicating mechanical information. It is language-agnostic while being a non-neutral name, which is appropriate: the name is a meta-label, not a system component. Deckpack remains valid for potential physical editions or commercial derivatives.

DC-02 — Base deck size: 52 + 2 Jokers

Decision: 54 cards total. **Rationale:** Jokers are structurally distinct in the standard deck — the only cards without a suit or rank. Excluding them

would waste a design resource. Advertising cards present in some decks are not standardized and are not part of the specification.

DC-03 — Permanent marking

Decision: annotations with permanent marker. **Alternatives considered:** erasable for all layers; mixed system. **Rationale:** permanent marking transforms each deck into a unique artifact. Variable states are tracked with physical tokens, which are already removable by nature. The mixed system is available as an optional practice for individual games but is not prescribed.

DC-04 — Open system: no rank → function mapping

Decision: DECK# does not assign mechanical functions by rank. **Discarded alternative:** cardpack-spec v0.2.0 had defined Space (A–6) / Resource (7–10) / Actor (J–Q–K) as a fixed partition. **Rationale:** that partition is the structure of a specific tactical game, not an open vocabulary. It blocks any game that wants to use an Ace as a character or a King as a territory. An open system does not prescribe what each card represents — each game built on top does that.

DC-05 — Seven primary glyphs (closed vocabulary)

Decision: exactly 7 glyphs: dot, line, cross, diagonal, triangle, circle, square. **Rationale:** they cover the primitive geometric shapes producible with 1–4 strokes. Sufficiently visually distinct from each other and from suits. The vocabulary is closed to guarantee interoperability — a glyph not in the list does not exist in DECK#. The asterisk from cardpack-spec v0.2.0 was excluded because visually ambiguous and motivated by a semantic role (Joker) that the specification must not pre-assign.

DC-06 — Glyph color: contrasting rule

Original decision (v0.1–v0.6): marker color is an independent semantic layer, orthogonal to the glyph; convention not prescribed. **Updated decision (v0.7):** glyph color follows a fixed contrasting rule — black glyphs on red cards (♥/♦), red glyphs on black cards (♣/♠). See DC-22. **Rationale for change:** a semantic axis that is “available but not prescribed” never enters the canonical annotation. The contrasting rule is simple, memorable, and guarantees high legibility on all cards without requiring a color column in the matrix.

DC-07 — Back: exception for quadrant markers

Original decision (v0.1–v0.5): back excluded from annotation. **Updated decision (v0.6):** the back of the 52 standard cards receives only quadrant markers. Two variants allowed: minimal (four short segments at midpoints) or extended (full + cross). The choice is left to the annotator. Jokers do not receive quadrant markers on the back. **Rationale for exception:** quadrant markers are identical on all 52 standard cards — they reveal no differential information. The original rationale (preserving anonymity for fog of war) does not apply to uniform markers. The benefit is eliminating the coexistence of quadrant markers and the primary glyph in S zone (DA-01 / DC-21, now superseded).

DC-08 — North pole convention

Decision: north pole = side with index in canonical position (NW at top left). Dot in SE as orientation marker. **Problem:** cards have vertical symmetry for ergonomic reasons. With spatial elements, this must be explicitly broken. **Rationale:** the SE dot is the least invasive solution — it covers nothing, is asymmetric, requires one stroke. A fixed arrow would have introduced spatial semantics even for games without orientation.

DC-09 — Symmetrical placement in hand mode

Decision: annotations in two mirror positions — primary glyph in S zone with mirror in N; modifier in NE corner with mirror in SW. See DC-18 for specific zones. **Source:** cardpack-spec v0.2.0 (zones updated in DC-18). **Rationale:** standard ergonomic convention for card decks — indices are

already duplicated and rotated for the same reason. Guarantees fan readability without additional rules.

DC-10 — Quadrant markers

Decision: present on the **back** of the 52 standard cards (migrated from front face in v0.6, see DC-07 rev.). Two variants: minimal (four short segments) or extended (+ cross). Absent on Jokers. **Source:** cardpack-spec v0.2.0 (universality established in DC-17; migration to back in DC-07 rev. v0.6). **Rationale:** on the front face, short segments were necessary to avoid covering pips. On the back, the extended variant (full cross) becomes viable without interference.

Revision v0.7.1 (2026-05-13): extended to Jokers on the back as well. See DC-11 rev. for rationale.

DC-11 — Jokers as uninstantiated variables

Decision: Jokers receive only the SE dot (north pole). No primary glyph, no modifier, no connectivity markers, no quadrant markers. **Discarded alternative:** fixed asterisk in cardpack-spec v0.2.0. **Rationale:** the asterisk assigns Jokers a specific functional role before any game uses them. An open system leaves Jokers as neutral surfaces. Their absence from the inherited axes is already sufficient to distinguish them.

Revision v0.7.1 (2026-05-13): Jokers receive quadrant markers on the back. The original rationale (Joker neutrality as a blank surface) applies to the front face, not the back: quadrant markers are uniform on all cards and convey no differential information. Their absence on Joker backs made them recognizable when face-down, compromising fog of war. The neutrality principle remains intact: the Joker is still free of pre-assigned semantics on the front face.

DC-12 — Base and Extended profiles

Decision: two profiles with increasing requirements. **Source:** cardpack-spec v0.2.0 (absent in DECK# v0.1). **Rationale:** the system must be accessible with minimal materials but should not pretend all games have the same requirements. The distinction is a scope declaration, not a hierarchy.

DC-13 — “Designer responsibilities” section

Decision: explicit list of what every game built on DECK# must specify.

Rationale: without this section, a designer does not know where their work begins. The list delimits the interface between specification and game.

DC-14 — CC BY 4.0 License

Decision: Creative Commons Attribution 4.0 International. **Alternatives considered:** CC0, CC BY-SA, proprietary license with usage permission.

Rationale: CC BY 4.0 balances openness and attribution. It permits commercial use, does not impose copyleft on derivatives, and requires only source citation. CC0 would have sacrificed any ecosystem traceability. CC BY-SA would have created friction for commercial games.

DC-15 — Single structural layer

Decision: glyphs are structural by definition. There is no separate semantic layer added by the game — the designer interprets the structural glyphs; they do not add their own layer. **Discarded alternative:** two-layer model (DECK# structural + game semantic). **Rationale:** the structural/semantic layer distinction is conceptual, not physical. Only one annotation layer exists on the deck. Semantics is a reading operation, not a writing one.

DC-16 — Three structural dimensions per card

Decision: each card is defined by three dimensions: Connectivity (open edge patterns across 15 configurations), Primary (glyph in S zone), Modifier (glyph in NE corner). Positional hierarchy, not semantic: identical Primary + Modifier (e.g., $\times/\times$) are legitimate combinations. **Rationale:** three independent dimensions create a combinatorial space ($15 \times 7 \times 7 = 735$ theoretical configurations) far richer than the 52 slots needed, guaranteeing design freedom in distribution.

DC-17 — Universal quadrant markers

Decision: quadrant markers are present on all 52 standard cards. Absent on Jokers. **Discarded alternative:** variable presence as a matrix dimension. **Rationale:** uniform spatial capability simplifies annotation and shifts capacity management to the game, which can ignore it, use it partially, or exploit it fully.

DC-18 — Zones: Primary in S, Modifier in NE

Decision: primary glyph in S zone (internal, lower middle edge) with mirror in N; modifier in NE corner with mirror in SW. Center not occupied by DECK# annotations. **Discarded alternative:** primary at center (blocked by pips on number cards). **Rationale:** S zone is visible when cards are held in a fan; it is clean and separated from pips for all cards. NE is the second most visible corner. The center remains readable for pips and available as a neutral visual space.

DC-19 — Annotation matrix as design artifact

Decision: the distribution of connectivity and glyphs across 52 cards is an authored artifact, orthogonal to rank/suit/color by deliberate construction. No algorithmic formula connects substrate properties to structural annotation. **Rationale:** formula-based orthogonality would require using rank/suit as an index, implicitly re-encoding them. Design-based orthogonality is more robust: no systematic correlation, controlled topological variety.

DC-20 — Pre-session annotation as absolute standard

Decision: the DECK# deck is fully annotated before any session. No in-game annotation for the structural layer. **Rationale:** consistent with the deck-as-fixed-artifact model. Removes a source of ambiguity from the “Designer responsibilities” section — it is not a game choice, it is a system constraint.

DC-22 — Glyph color: contrasting rule

Decision: black glyphs on red cards (♥/♦); red glyphs on black cards (♣/♠). Fixed rule, no color column in the matrix. **Discarded alternatives:** (1) color as independent semantic axis in the matrix — requires designed distribution and adds complexity; (2) all glyphs in black — loses legibility on black cards. **Rationale:** contrasting color is entirely derivable from suit (zero new differential information) but maximizes visual separation between annotation and substrate. The rule's simplicity is its primary value: anyone annotating the deck can derive it without consulting the matrix.

DC-21 — Coexistence of quadrant marker and primary glyph in S zone

SUPERSEDED in v0.6 by DC-07 rev. Quadrant markers migrate to the back: coexistence with the primary glyph in S zone is no longer an issue. Section 5.1 of the spec (placement diagram) has been removed.

Open decisions

DA-01 — CLOSED (DC-16, DC-18): coexistence of multiple glyphs in the same zone is not an issue — each zone holds at most one DECK# glyph.

DA-02 — Deck subset notation: standard system for documenting which subset is used (e.g., DECK#/**♠**A-6).

DA-03 — CLOSED: compatibility with non-standard decks is out of scope by definition. DECK#’s prerequisite is a standard French/poker deck (52 + 2 Jokers). Spanish, German, and tarot decks do not satisfy the inherited substrate of section 3.

DA-04 — CLOSED (rejected): quadrant markers on the back would be useful in tile mode with a face-down card, but the back pattern varies from deck to deck in unpredictable ways — on decks with dense patterns the markers would be illegible or hidden. DC-07 (back not annotatable) remains valid without exceptions.

DECK#

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